



BACKLOG SUBJECT EXAMINATIONS - FEBRUARY 2024

B.E. - Artificial Intelligence and Data		
Program	: Science / Artificial Intelligence and Machine Learning	Semester : III
Course Name	: Data Structures	Max. Marks : 100
Course Code	: AD33 / AI33	Duration : 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Explain dynamic memory allocation function with suitable examples. CO1 (08)
 - Compute the failure functions for the following patterns using KMP algorithm. CO1 (04)
 - aabaaabab
 - abbabbaba.
 - Write necessary C functions to add two polynomials represented using an array. CO1 (08)
- Define an algorithm. Explain its various characteristics. CO1 (04)
 - Write a C function for pattern matching by checking end indices first. CO1 (08)
 - Explain Self-Referential Structures with an example. CO1 (08)

UNIT - II

- Define stack. Illustrate basic operations on stack with suitable diagrams. CO2 (06)
 - Write a C function to display a circular queue implemented as an array. CO2 (07)
 - Write a C function to compare contents of two stacks using array implementation. CO2 (07)
- Write necessary C functions to evaluate a postfix expression using stack. CO2 (06)
 - Describe limitations of a linear queue with suitable examples. CO2 (08)
 - Illustrate operations on multiple stacks with a neat diagram. CO2 (06)

UNIT - III

- What is Doubly Linked List? Write an algorithm to insert and delete a node in at the end of a Doubly Linked List. CO3 (10)
 - Use the postfix evaluation algorithm to evaluate the following expressions. Assume A=1, B=2, C=3. CO3 (10)

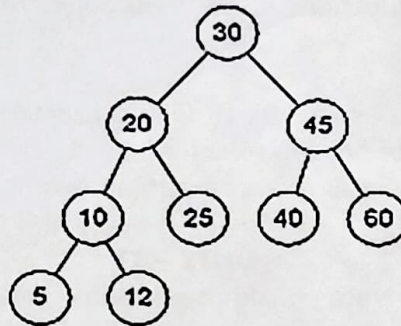
Tabulate the Results using Stack

 - AB+C-BA+C\$-
 - ABC+*CBA-+

6. a) Write a C function that deletes a node containing a number NUM from a circular linked list. CO3 (06)
b) Illustrate a linked stack and its operations using neat diagrams. CO3 (08)
c) Write a C function to erase all the terms in a polynomial represented as a singly linked list. CO3 (06)

UNIT – IV

7. a) Show the following keys as the list representation of a tree. CO4 (06)
(A(B(E(K,L),F),C(G),D(H(M),I,J)))
b) Write a recursive C function to search for an element in a binary search tree. CO4 (08)
c) Write an ADT for a PriorityQueue and include necessary operations. CO4 (06)
8. a) Define with an example each of the following: CO4 (06)
i) Binary tree ii) Full binary tree iii) Complete binary tree.
b) Describe satisfiability problem. Justify how it can be solved using binary tree with a suitable example. CO4 (08)
c) Write the Pre-Order, In-Order and Post-Order traversals of the following binary tree. CO4 (06)



Binary Search Tree

UNIT – V

9. a) Explain Breadth first search of a graph using a suitable example. CO5 (07)
b) Demonstrate the construction of a multilists for a graph with an example. CO5 (07)
c) Justify the need for an AVL tree over unbalanced binary trees. CO5 (06)
10. a) Write a C function to create a graph using an adjacency matrix representation. CO5 (06)
b) Write an ADT for a Graph and include necessary operations. CO5 (08)
Explain the relation between degree, vertices and edges in a directed graph with a suitable example. CO5 (06)
